MARYAM REHAN

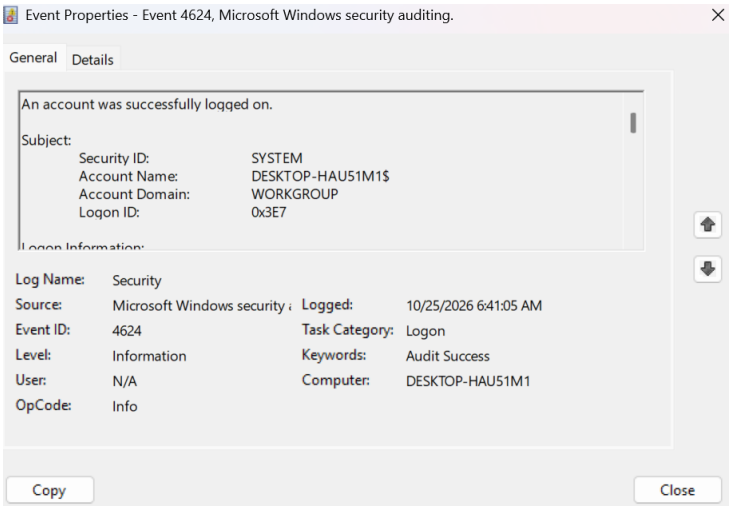
INTERNSHIP REPORT CYBER SECURITY (SOC)

Week 2 — Task 4: Windows Logging and Event Viewer

Objective

The objective of these tasks was to develop practical knowledge of Linux system administration, log analysis, and Windows security investigation from a Security Operations Center (SOC) perspective. The Linux task focused on learning essential commands for navigating directories, creating and viewing files, searching log entries, and understanding system logs stored in locations such as /var/log. The Windows investigation task focused on understanding system information, user logon activity, administrative privileges, suspicious scheduled tasks, network ports, and potential indicators of compromise. These activities aimed to strengthen my ability to examine system records, recognize unusual behavior, understand security events, and support threat detection and incident response in Linux and Windows environments.

Exploring Windows Security Logs



Event ID 4624 Successful Logon

Description:

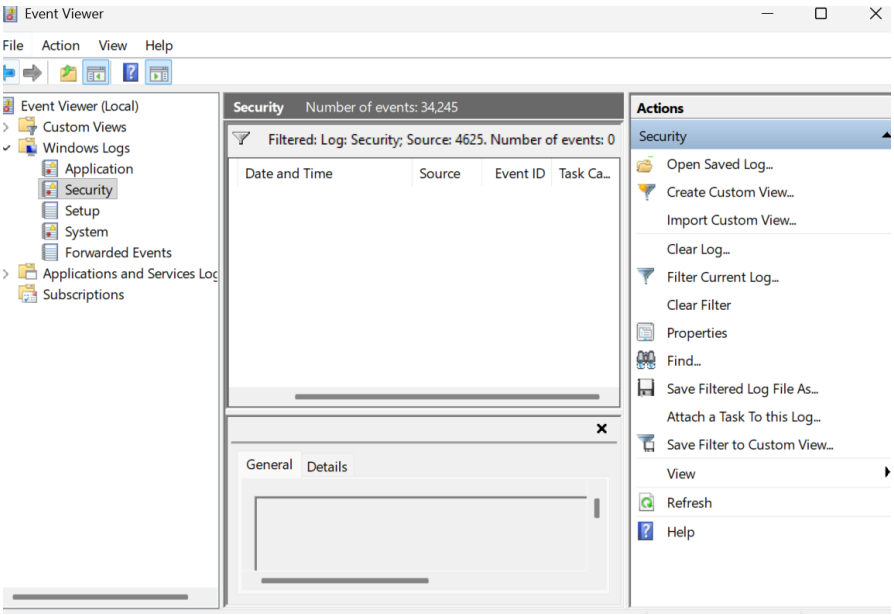
Event ID 4624 is generated when a user successfully logs on to a Windows system. It records useful information such as the account name, logon type, date and time, and other authentication details. SOC analysts monitor this event to identify unusual login activity and investigate potentially unauthorized access.

Observation:

The Windows Security log was filtered to locate Event ID 4624. The event indicates a successful logon, and its available details can help an analyst understand when and how the login occurred.

Security Relevance:

SOC analysts can review successful logon events to identify unexpected login times, unfamiliar accounts, or unusual logon types. Event ID 4624 should be correlated with other security events before concluding that an activity is suspicious.

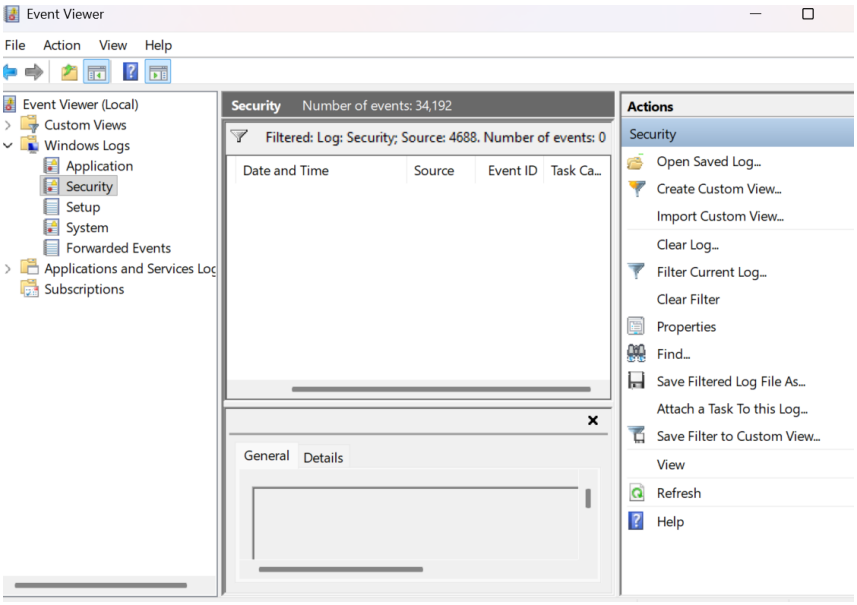


Event ID 4625 Failed Logon Investigation

Observation: The Windows Security log was filtered for Event ID 4625. The filter returned zero matching events in the available log.

Interpretation: No matching failed-logon event was available for analysis in the selected log. This does not prove that failed logon attempts never occurred; the relevant events may not have been recorded or may no longer be available.

Status: Practical analysis pending because no matching Event ID 4625 was found.

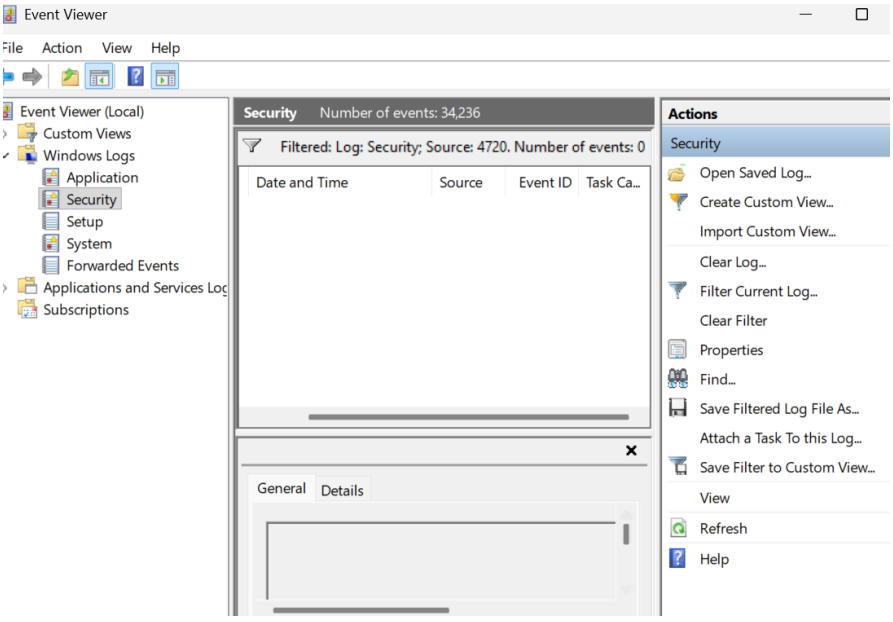


Event ID 4688 Process Creation

Observation: The Security log was filtered for Event ID 4688, which records process creation when the relevant auditing is enabled. The filter returned zero matching events in the available log.

Analysis: No matching process-creation event was available for examination. This may relate to the audit configuration or the available log contents; the exact cause has not been verified.

Status: Event-level practical verification pending.



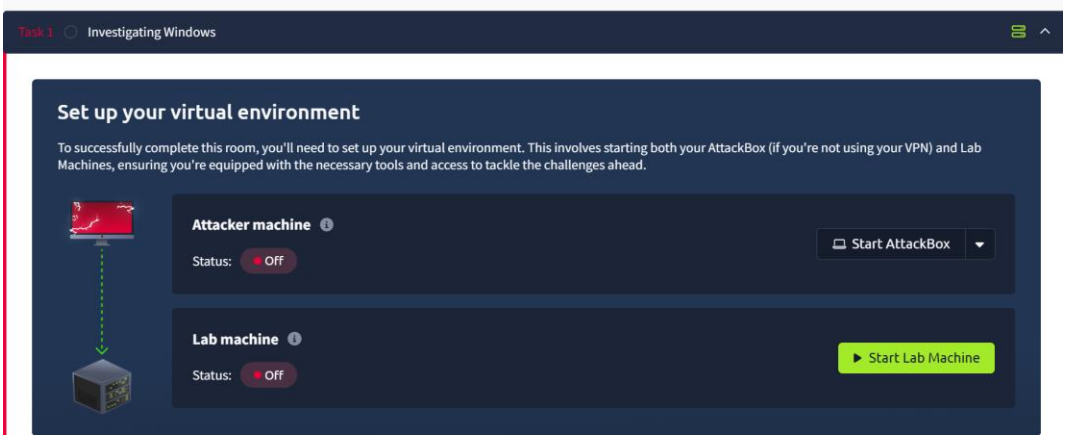
Event ID 4720 — User Account Created

Description: Event ID 4720 is generated when a new user account is created on a Windows system. This event records information about the newly created account and the account responsible for creating it.

Security Relevance: SOC analysts monitor this event to identify unauthorized user account creation. An unexpected account may indicate suspicious activity or an attempt to gain persistent access to a system. Analysts should verify whether the account creation was authorized.

Observation: The Windows Security log was filtered for Event ID 4720. No matching events were found in the available log entries.

Try Hack Me Learning Activities



Whats the version and year of the windows machine?

Windows Server 2016 — ✓ Correct Answer

Which user logged in last?

Administrator — ✓ Correct Answer

When did John log onto the system last?

Answer format: MM/DD/YYYY H:MM:SS AM/PM

03/02/2019 5:48:32 PM — ✓ Correct Answer

What IP does the system connect to when it first starts?

10.34.2.3 — ✓ Correct Answer

Answer format: List them in alphabetical order.

Guest, Jenny ✓ Correct Answer

Whats the name of the scheduled task that is malicous.

Clean file system ✓ Correct Answer

What file was the task trying to run daily?

nc.ps1 ✓ Correct Answer

What port did this file listen locally for?

1348 ✓ Correct Answer

When did Jenny last logon?

Never ✓ Correct Answer

At what date did the compromise take place?

Answer format: MM/DD/YYYY

03/02/2019 ✓ Correct Answer

During the compromise, at what time did Windows first assign special privileges to a new logon?

Answer format: MM/DD/YYYY HH:MM:SS AM/PM

03/02/2019 4:04:49 PM ✓ Correct Answer

What tool was used to get Windows passwords?

Mimikatz ✓ Correct Answer

What was the attackers external control and command servers IP?

76.32.97.132 ✓ Correct Answer

What was the extension name of the shell uploaded via the servers website?

.jsp ✓ Correct Answer

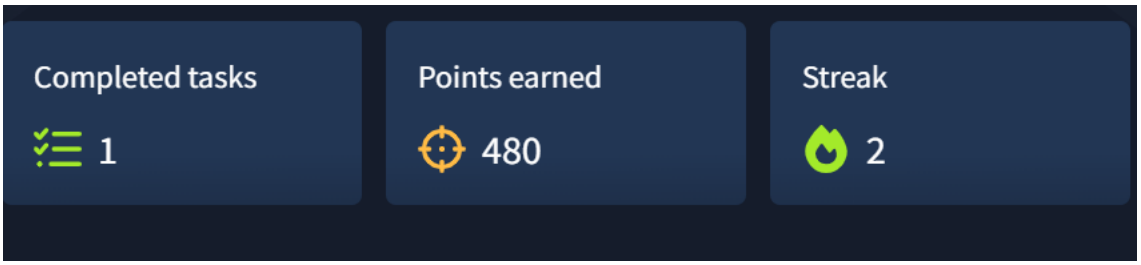
What was the last port the attacker opened?

1337 ✓ Correct Answer

Check for DNS poisoning, what site was targeted?

google.com ✓ Correct Answer

Room completed: 100%



Task 1: Investigating Windows

Detailed Description:

The objective of this task was to understand how to investigate a Windows system from a cybersecurity and Security Operations Center (SOC) perspective. Windows systems generate useful information about user activities, system operations, application behavior, scheduled tasks, and network connections. Analyzing this information helps security analysts understand normal system activity and identify suspicious behavior that may indicate a security incident.

During this task, the investigation covered important areas of Windows security, including user logon activity, administrative accounts, scheduled tasks, running files, network ports, and potential indicators of compromise. Windows event logs are particularly useful for reviewing authentication attempts, privilege assignments, process activity, and other security-related events. These records can help analysts reconstruct a sequence of events and investigate how an attacker may have accessed or interacted with a system.

The task also introduced the importance of identifying suspicious scheduled tasks, examining files executed automatically, checking locally listening ports, and recognizing possible command-and-control (C2) communication. In addition, DNS poisoning was considered as a potential attack technique that can redirect users to unintended websites.

This practical learning activity helped develop an understanding of Windows investigation procedures and the types of evidence that SOC analysts may examine when responding to security alerts. It also highlighted the importance of collecting evidence carefully, correlating relevant event records, and verifying findings before drawing conclusions.

Learning Outcomes:

- Understand the purpose of Windows system investigation.
- Recognize the importance of Windows event logs in security monitoring.
- Identify user accounts and review logon-related information.
- Understand the security risks associated with suspicious scheduled tasks and executable files.

- Recognize the role of ports, network connections, and external C2 addresses in incident investigations.
- Understand how indicators of compromise can support incident detection and analysis.
- Develop a structured approach to investigating suspicious Windows activity.

Task Status: Completed in TryHackMe, as indicated by the green checkmark shown beside the task heading.

Conclusion

In conclusion, the Linux Logs and Investigating Windows tasks provided valuable foundational knowledge of system monitoring, log analysis, and security investigation in Linux and Windows environments. Through the Linux task, I learned to use essential terminal commands, navigate directories, manage files, and understand the importance of system logs for troubleshooting and security monitoring. Through the Windows investigation task, I gained an understanding of user logon activity, administrative privileges, suspicious scheduled tasks, network ports, and potential indicators of compromise. Both tasks highlighted the importance of analyzing system records, identifying unusual activities, and verifying evidence before drawing conclusions. Overall, these practical exercises strengthened my understanding of SOC operations, security event investigation, and the role of log analysis in detecting and responding to potential security incidents.